The truth is that deep learning constitutes just a part of machine learning gathering all those models built with *deep* (intended as “with many layers”) neural networks, and machine learning in turn is just a branch of artificial intelligence.

In particular, machine learning is the branch that is focused on developing mathematical models that describe a system after learning its characteristics from data.

ML and deep DL can achieve impressive, eye-catching, and sometimes incredible results (at the time of writing you can think of, for instance, how life-like artificially-generated faces, landscapes and even movies created by GANs[2] look like), but they can’t, and neither they aim to, build an “intelligent” agent, something closer to the romantic idea of an artificial conscious being that cult movies like *Short Circuit* or *War Games* gave us: that is rather the goal of *general artificial intelligence*.

12.1.1 Types of Learning

The main classification of machine learning models is based on the type of “learning” they perform, in particular focusing on the way we provide feedback to the model during training:

- **Supervised learning**: these models are trained on labeled data, i.e. for each entry in the training set, there is a label (for *classification* algorithms) or a value (*regression* algorithms) that is associated to that entry; training will tune models’ parameters in order to